

Byeong Gwon Lee

3D Computer Vision Researcher

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3D Reconstruction · Gaussian Splatting · SLAM · Multi-View Stereo

EXPERIENCE

KETI

Researcher

Sep. 2026 - Present

StradVision

Researcher

Jul. 2026 - Jul. 2026

MotifDrive

Researcher

May 2026 - Jul. 2026

RTM AI Image

2D Computer Vision Researcher

Sep. 2022 - Feb. 2024

Manufacturing defect detection, image segmentation, model optimization, and few-shot auto labeling. Progressed from research intern to data scientist.

EDUCATION

Dongguk University

M.S. in Computer Science and Artificial Intelligence | GPA: 4.5 / 4.5

Advisor: Prof. Soohwan Song

Thesis: Online 3D Gaussian Splatting Modeling with Novel View Selection

Mar. 2024 - Feb. 2026

Hanyang University ERICA

B.S. in Software Engineering | GPA: 3.68 / 4.5

Mar. 2020 - Feb. 2023

PUBLICATIONS - PUBLISHED / ACCEPTED

Online 3D Gaussian Splatting Modeling with Novel View Selection

Byeonggwon Lee, Junkyu Park, Khang Truong Giang, Soohwan Song

IJCAI 2025 | Published | First Author

Paper

MVS-GS: High-Quality 3D Gaussian Splatting Mapping via Online Multi-View Stereo

Byeonggwon Lee, Junkyu Park, Khang Truong Giang, Sungho Jo, Soohwan Song

IEEE Access | Published | First Author

Paper

Stereo-GS: Online 3D Gaussian Splatting Mapping Using Stereo Depth Estimation

Junkyu Park, Byeonggwon Lee, Sanggi Lee, Soohwan Song

MDPI Electronics | Published | Co-first Author

Paper

MANUSCRIPTS - UNDER REVIEW

The following manuscripts are under review and have not been accepted for publication.

M2Depth: Unifying Monocular Depth Foundation Priors with Multi-View Stereo

Byeonggwon Lee, Sanggi Lee, Siwoo Lee, Khang Truong Giang, Soohwan Song

IEEE Transactions on Multimedia (TMM) | Under Review | First Author

arXiv

RESEARCH PROJECTS

Online/Offline Real-Time 3D Reconstruction Using Drone Imagery

Oct. 2024 - Apr. 2025

Mobility One Inc. | Core researcher

Combine SLAM camera tracking with multi-view stereo in an online/offline reconstruction pipeline.

Adaptive keyframe selection, hierarchical bundle adjustment, and loop closure for pose drift correction.

Completed delivery to Mobility One and secured a follow-up project.

3D Reconstruction with Diffusion-based Inpainting

Jun. 2025 - Dec. 2025

Mobility One Inc. | Core researcher

Use COLMAP and OpenMVS with diffusion-based image interpolation and inpainting.

A geometric constraint filter checks generated imagery for consistency before reconstruction.

Reconstructed missing mesh regions without additional image capture.

PATENTS & PROFESSIONAL SERVICE

System and Method for 3D Modeling Using Online Multi-view Stereo

Korean patent application and PCT international patent application. Principal inventor.

IEEE Access - Peer Reviewer | Nov. 2025 - Present

AWARDS & RECOGNITION

Dongguk University graduate school fair - Outstanding Research Award, 2025.

Research featured in the Dongguk University graduate school newsletter, 2025.

M.S. GPA: 4.5 / 4.5 throughout the degree, 2024 - 2026.

TECHNICAL SKILLS

Programming: Python, PyTorch, C/C++, CUDA, Shell scripting.

3D vision: Gaussian Splatting, Multi-View Stereo, SLAM, depth estimation, bundle adjustment, COLMAP, OpenMVS.

2D vision: Object detection, image segmentation, keypoint detection, few-shot learning, defect inspection.

Infrastructure: Linux, Git, Docker, Kubernetes.